LESSON 7.6

# Determine Distance on the Coordinate Plane

**Start here** · Words, worked example, and sentence starters

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can find the distance between two points on the coordinate plane using absolute value.

**WHY THIS MATTERS** Captain Vega has two treasure locations on her map: a buried chest at (-3, 2) and a hidden cave at (4, 2). Both are at the same height (y = 2) on the map. She needs to figure out how far apart they are so she knows how much rope to bring. Another pair of treasures is at (1, -3) and (1, 5) — same column, different rows. How far apart is each pair?

START HERE

## Words for this lesson

Read each word before you begin. Say it out loud.

| WORD | WHAT IT MEANS | EXAMPLE |
|---|---|---|
| **Distance** *Spanish: Distancia* | How far apart two points are. It is never negative. | always a positive length |
| **Absolute value** *Spanish: Valor absoluto* | How far a number is from zero. It is never negative. | $\|-7\| = 7$ |
| **Horizontal distance** *Spanish: Distancia horizontal* | How far apart two points are going left or right. | uses the x-coordinates |
| **Vertical distance** *Spanish: Distancia vertical* | How far apart two points are going up or down. | uses the y-coordinates |
| **Coordinate plane** *Spanish: Plano cartesiano* | A grid with a line going across and a line going up to plot points. | x-axis across, y-axis up |
| **Integer** *Spanish: Número entero* | Whole numbers and their opposites, like -2, -1, 0, 1, 2. | ..., -3, -2, -1, 0, 1, 2, 3, ... |

HOW IT WORKS

## Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

**Each unit is 1 meter. Find the distance from P(−9, 4) to Q(3, 4), and from P to R(−9, 11).**

1. P and Q share y = 4, so use x. −9 and 3 have opposite signs: 9 + 3 = 12 m.
2. P and R share x = −9, so use y. 4 and 11 are both positive: 11 − 4 = 7 m.

**Answer:** PQ = 12 m and PR = 7 m

NOW YOU TRY

## Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

**Now let's find the distance from (2, 3) to (2, 8). What do these points share? The same x = 2, so it is a vertical line.**

1. _____
2. _____
3. _____

**Answer:** _____

SAY IT AND WRITE IT

## Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The two points share the _____ -coordinate, so the segment runs _____ .
- I used absolute value because _____ .
- The points sit on _____ sides of the axis, so I _____ the two distances.
- The distance is _____ units, which means _____ on this map.

**Word bank** distance | absolute value | horizontal | vertical | coordinate plane | axis | units | add | subtract | x-coordinate | y-coordinate | segment

WATCH OUT!

## A common mistake

A common mistake in Distance on the Coordinate Plane is subtracting the coordinates as if both points were on the same side of zero, instead of adding their absolute values when the points are on opposite sides. For example, to find the distance between (-6, 3) and (2, 3), a student might subtract 6 − 2 = 4, but since -6 and 2 are on opposite sides of zero, the absolute values must be added instead: |-6| + |2| = 6 + 2 = 8. The correct distance is 8 units, not 4. Before you submit, ask: 'Are these two points on opposite sides of zero

LESSON 7.6

# Determine Distance on the Coordinate Plane

**Version A** · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can find the distance between two points on the coordinate plane using absolute value.

**WHY THIS MATTERS** Captain Vega has two treasure locations on her map: a buried chest at (-3, 2) and a hidden cave at (4, 2). Both are at the same height (y = 2) on the map. She needs to figure out how far apart they are so she knows how much rope to bring. Another pair of treasures is at (1, -3) and (1, 5) — same column, different rows. How far apart is each pair?

## PART 1 Understand the idea

models and meaning

**1** **Complete the table. Show how you found each value.**

Both points in each row share a coordinate. Find the distance.

| Point 1 | Point 2 | Shared Coordinate | Distance |
|---|---|---|---|
| (-5, 1) | (3, 1) | y = 1 | |
| (2, -4) | (2, 6) | x = 2 | |
| (-3, -2) | (-3, 4) | x = -3 | |
| (-1, 5) | (4, 5) | y = 5 | |

**HINT** Each row shares a coordinate, so every distance is a straight horizontal or vertical count.

SHOW YOUR WORK

**2** **Mark and label each value on the number line.**

Plot -4 and 2 on the number line. Count the distance between them.




**HINT** Distance is the number of unit hops between the two points — direction doesn't matter.

SHOW YOUR THINKING

## PART 2 Apply it

real-world problems

**3** **Circle the letter of the best answer. Show how you know.**

What is the distance between (-4, 3) and (2, 3)?

(A) 2 units
(B) 4 units
(C) 6 units
(D) 8 units

**HINT** Both points have y = 3, so they sit on the same horizontal line — only the x-values differ.

SHOW YOUR WORK

**4** **Circle the letter of the best answer. Show how you know.**

What is the distance between (5, -2) and (5, 4)?

(A) 2 units
(B) 4 units
(C) 6 units
(D) 10 units

**HINT** Both points share x = 5 — the segment is VERTICAL, so compare the y-values -2 and 4.

SHOW YOUR WORK

**5** **Circle the letter of the best answer. Show how you know.**

A rectangle has vertices at (1, 1), (5, 1), (5, 4), (1, 4). What is its perimeter?

(A) 8 units
(B) 12 units
(C) 14 units
(D) 20 units

**HINT** Perimeter is the distance all the way around — you need the width AND the height first.

**Formula** _____

**Put the numbers in** _____

**Work it out** _____

**Answer with its unit** _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

## PART 3 Explain and correct

reasoning

**6** **Find the mistake. Explain it, then show the correct work.**

### Spot the Common Mistake

1. **Problem** Find the distance between (2, 1) and (2, 6).
2. **Student thinking** Both share x = 2, so add the absolute values: |1| + |6| = 1 + 6 = 7.
3. **Student answer** The distance is 7 units.

**Which step has the mistake? Explain what went wrong, then show the correct work.**

**HINT** Are 1 and 6 on the same side of zero or opposite sides?

CORRECT WORK

**Correct answer:** _____

EXPLAIN YOUR THINKING

**How did you use coordinates and absolute value to find the distance between two points in the same row or column?**

**SENTENCE STARTER** The points ___ and ___ share the same ___ -coordinate. To find the distance, I found |___| + |___| = ___, which equals ___ units.

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.6

# Determine Distance on the Coordinate Plane

**Version B** · Core practice

**MASTERY CHECK**
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can find the distance between two points on the coordinate plane using absolute value.

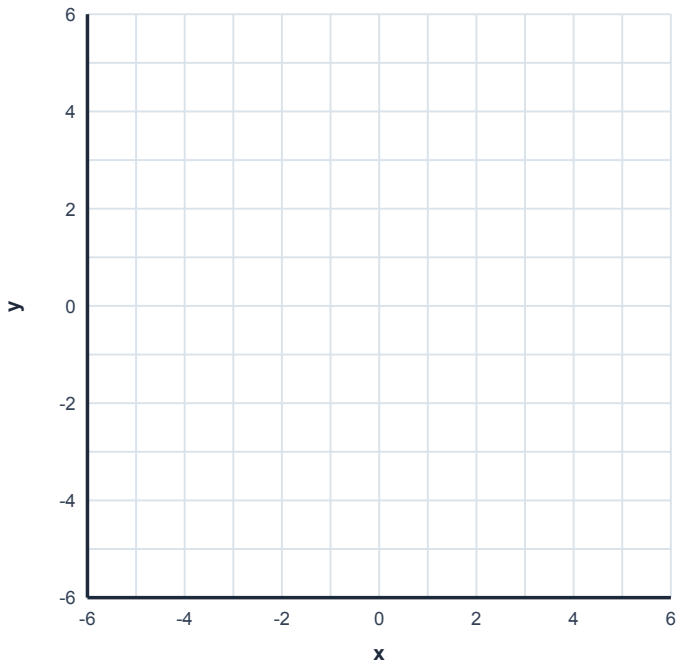
**WHY THIS MATTERS** Captain Vega has two treasure locations on her map: a buried chest at (-3, 2) and a hidden cave at (4, 2). Both are at the same height (y = 2) on the map. She needs to figure out how far apart they are so she knows how much rope to bring. Another pair of treasures is at (1, -3) and (1, 5) — same column, different rows. How far apart is each pair?

## PART 1 Understand the idea

models and meaning

**1** **Plot and label each point on the grid.**

Plot these pairs and find each distance: A(-5, 3) to B(2, 3), and C(4, -2) to D(4, 4)




**SHOW YOUR THINKING**

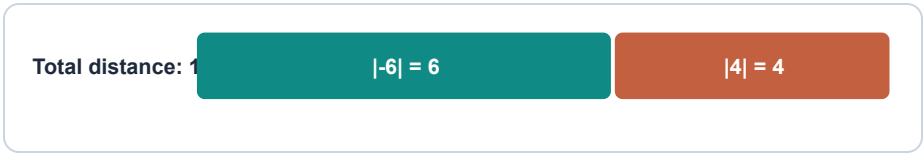
**2** **Complete the table. Show how you found each value.**

Find the perimeter of each rectangle using its vertex coordinates.

| Rectangle | Vertices | Width | Height | Perimeter |
|---|---|---|---|---|
| Rect 1 | (-3, 1), (4, 1), (4, 5), (-3, 5) | | | |
| Rect 2 | (-2, -4), (6, -4), (6, 2), (-2, 2) | | | |

SHOW YOUR WORK

**3** **Use the model to solve. Show your work.**




What is the total distance from (-6, 3) to (4, 3)?

SHOW YOUR WORK

**Answer:** _____

## PART 2 Apply it

real-world problems

**4** **Circle the letter of the best answer. Show how you know.**

Two points are at (-3, -7) and (-3, 5). What is the distance between them?

(A) 2 units
(B) 7 units
(C) 8 units
(D) 12 units

SHOW YOUR WORK

**5** **Circle the letter of the best answer. Show how you know.**

What is the distance between (2, 3) and (2, 8)?

(A) 5 units
(B) 5.5 units
(C) 6 units
(D) 11 units

SHOW YOUR WORK

## PART 3 Explain and correct

reasoning

**6** **Find the mistake. Explain it, then show the correct work.**

### Spot the Common Mistake

1. **Problem** Find the distance between (-5, 2) and (3, 2).
2. **Student thinking** Both share y = 2, so subtract the x-values: 5 - 3 = 2.
3. **Student answer** The distance is 2 units.

**Which step has the mistake? Explain what went wrong, then show the correct work.**

CORRECT WORK

**Correct answer:** _____

EXPLAIN YOUR THINKING

**How did you use coordinates and absolute value to find the distance between two points in the same row or column?**

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.6

# Determine Distance on the Coordinate Plane

**Challenge** · Extension

**MASTERY CHECK**

- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can find the distance between two points on the coordinate plane using absolute value.

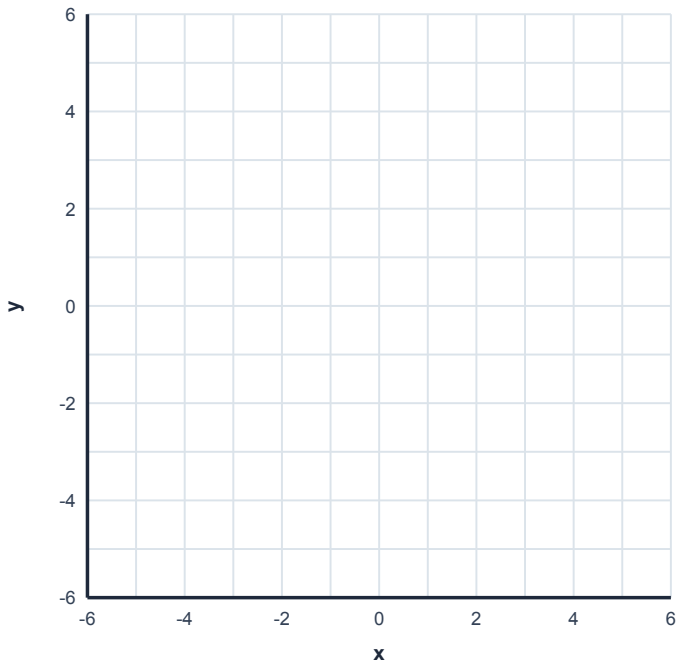
**WHY THIS MATTERS** Captain Vega has two treasure locations on her map: a buried chest at (-3, 2) and a hidden cave at (4, 2). Both are at the same height (y = 2) on the map. She needs to figure out how far apart they are so she knows how much rope to bring. Another pair of treasures is at (1, -3) and (1, 5) — same column, different rows. How far apart is each pair?

## PART 1 Understand the idea

models and meaning

**1** **Plot and label each point on the grid.**

Plot a rectangle with one vertex in each quadrant. Label the vertices and calculate the perimeter.




**SHOW YOUR THINKING**

## PART 2 Apply it

real-world problems

**2** **Circle the letter of the best answer. Show how you know.**

A rectangle has corners (1, 2), (1, 5), (5, 5), and (5, 2). What is its perimeter?

(A) 7 units

(B) 10 units

(C) 12 units

(D) 14 units

**Formula** _____

**Put the numbers in** _____

**Work it out** _____

**Answer with its unit** _____

**3** **Circle the letter of the best answer. Show how you know.**

How far apart are the points (-2, 5) and (-2, -3)?

(A) -8 units

(B) 2 units

(C) 8 units

(D) 15 units

SHOW YOUR WORK

## PART 3 Explain and correct

reasoning

**4** Find the mistake. Explain it, then show the correct work.

### Find the Distance Error

1. **Problem** Find the distance between (-3, 4) and (5, 4)
2. **Student thinking** Subtract: 5 - 3 = 2
3. **Student answer** The distance is 2 units

**Which step has the mistake? Explain what went wrong, then show the correct work.**

CORRECT WORK

**Correct answer:** _____

**5** Answer in complete sentences.

A rectangle has vertices at (-4, -2), (3, -2), (3, 5), and (-4, 5). Find the length, width, perimeter, and area of the rectangle. Show your work using absolute value.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

**How did you use coordinates and absolute value to find the distance between two points in the same row or column?**

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help